Kevin Tan

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Education

California Polytechnic State University, San Luis Obispo Sept *2021 - June 2025*

B.S. Computer Science GPA: 3.732

Relevant Coursework: Data Structures, Database Design/Modeling/Implementation, Data Science, Data Visualization, Data Mining, Machine Learning, Deep Learning, Natural Language Processing, Distributed Computing, Dynamic Web Development

Technical Skills

Python (Pandas, NumPy, PyTorch), Java, C, C#, Go, JavaScript (Vega-Lite), TypeScript (React), HTML, CSS, SQL (MySQL), NoSQL (MongoDB), Git, Unix/Linux, Docker, Unit Testing, Microsoft Word/Excel/PowerPoint

Work Experience

Infios June 2025 - August 2025

Software Engineering Intern Remote

- Built a Markdown document chunking system optimized for **retrieval-augmented generation (RAG)**, enabling intelligent query-to-content matching and dynamic response generation based on user intent
- Set up pipelines to add **Docker** Images to the **Oracle Cloud Infrastructure (OCI)** and create Helm charts to host the chatbot on a **Kubernetes** cluster

Bendix Commercial Vehicle Systems June 2023 - Sept 2023

Software Engineering Intern Irvine, CA

- Developed an automated response handler in **C#** to deal with validation, activation, and deactivation of on and off-boarded vehicles using data provided by the original equipment manufacturer
- Utilized **SQL** queries to write a binary classification algorithm for an automatic event ranking system to evaluate driver safety based on dozens of data points during different points of a trip

Related Experience

Oracle Lens - League of Legends Esport Statistics Visualizer Jan 2025 - June 2025

Fullstack & Deep Learning Cal Poly SLO

- Designed and implemented a **SQL** database and **RESTful API** to serve data on teams, players, and game statistics spanning 36,000+ matches over 11 years
- Developed and deployed a full-stack web application using **TypeScript (React, Node.js)** to visualize tournament insights and player/team performance metrics
- Built a machine learning pipeline using **long short term memory recurrent neural networks (LSTM RNNs)** to model team performance trends and a neural network to predict match outcomes, achieving a 69% F1 score

Solving Where's Waldo with Spatial Partitioning May 2025 - June 2025

Distributed Computing Cal Poly SLO

- Fine-tuned a pretrained **PyTorch** vision transformer model to classify 64x64 images containing Waldo or not with an accuracy of 98.91% and an F1 score of 91.61%
- Implemented a gossip heartbeat protocol for a leader and worker distributed system in **Go** where the leader created image partitions while the workers classified the images

Analysis of Walkability throughout the United States May 2025 - June 2025

Data Visualization Cal Poly SLO

- Analyzed walkability across U.S. block groups and counties to identify trends distinguishing highly walkable areas from less walkable ones
- Built interactive visualizations using **Vega-Lite** and **D3**, including choropleths and regression plots, to enable exploratory analysis and intuitive data insights

League of Legends Worlds 2024 Draft Exploration May 2025

Data Visualization Cal Poly SLO

- Developed an interactive **JavaScript** visualization using **Vega-Lite** to illustrate shifts in the draft metagame across tournament phases
- Analyzed trends by comparing average draft position (picks and bans) against win rates to reveal fluctuations in champion priority and performance over time

Mel Spectrogram Music Genre Classification Feb 2025 - Mar 2025

Deep Learning Cal Poly SLO

- Augmented 1,000 music samples across 10 genres using pitch shifting, noise injection, and segment reordering, then converted them into mel spectrograms for model training
- Designed and trained a custom eight layer convolutional neural network (CNN) using **PyTorch** to classify music samples by genre, achieving 73.7% accuracy and an average F1-score of 73.4%

Market Trends Analysis in California Commercial Real Estate April 2025

Data Science DataFest

- Developed and trained Random Forest models in **Python (scikit-learn)** on commercial leasing transaction data (2018–2024) from U.S. markets to predict rent values and perform feature importance analysis
- Analyzed feature importance from the Random Forest models, identifying walkability, company size, revenue, leased square footage, location type, and seasonal timing as top predictors of rent, achieving a cross-validated Root Mean Squared Error of 0.85

Exploration of the Effects of Popularity on Spotify Song Classifications Feb 2025 - Mar 2025

Data Science Cal Poly SLO

- Trained four K-Means clustering models in **Python (scikit-learn)** to group songs by predicted genre using audio features, with and without popularity metrics, revealing a 45.6% average difference in classifications and suggesting bias in user-perceived genre consistency

Detecting Names and Self-Identifying Speakers from Legislative Speech Nov 2024 - Dec 2024

Natural Language Processing Cal Poly SLO

- Extracted and labeled over 10,000 examples of names and self-identifying speakers from a dataset of 70,000 statements made during 535 legislative sessions in Digital Democracy’s legislative speech database
- Optimized **spaCy**’s English transformer pipeline for named entity recognition and fine-tuned the model to focus on identifying only names and self-identifying speakers, resulting with a model with 92.92% precision and 93.45% recall